

State of Developer Ecosystem 2024: Language Trends, Satisfaction, and Strategic Shifts

A leadership lens on how usage is changing, what developers prefer, and where language choices map to application domains.

Based on JetBrains' annual survey of 23,262 developers worldwide covering programming language usage, tooling preferences, AI sentiment, and employment trends.

language_shift()

usage → preference

domain → strategy

Python closes gap

TypeScript wins DX

Rust + Go grow infra



Python Closes the Gap: 8 Years of Language Usage Shifts

JavaScript remains most used, but Python has nearly closed the gap; TypeScript tripled while PHP contracted.

Language	2017	2020	2022	2024	Trend
 JavaScript	65%	70%	65%	61%	Declining from peak
 Python	32%	55%	53%	57%	Steady growth
 HTML/CSS	60%	61%	54%	51%	Gradual decline
 SQL	42%	56%	49%	48%	Plateaued
 Java	47%	54%	48%	46%	Slow decline
 TypeScript	12%	28%	34%	37%	Strong growth
 Go	8%	19%	19%	18%	Plateaued after growth
 PHP	30%	27%	20%	17%	Significant decline
 Rust	--	7%	9%	11%	Steady growth
 Kotlin	2%	17%	16%	14%	Plateaued

LEADERSHIP READ

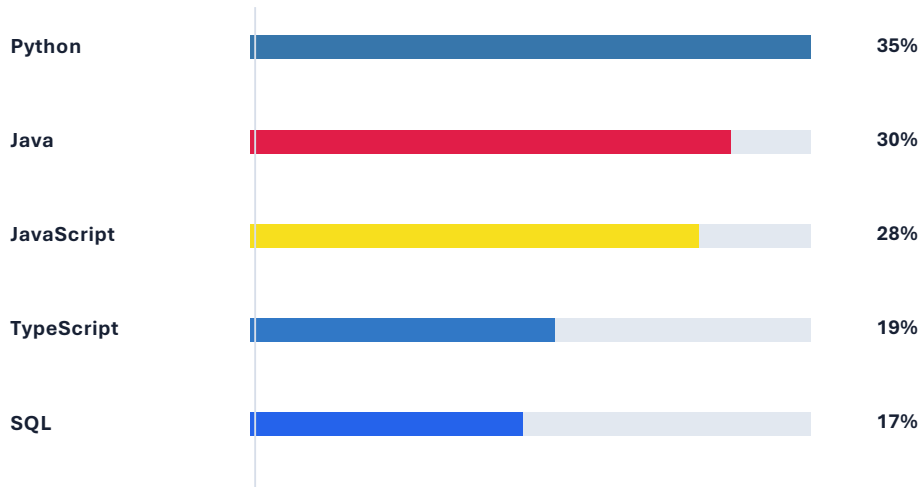
Top-5 reshuffle, not wholesale replacement

-  **JavaScript:** still leads overall at 61%, down from a 70% peak.
-  **Python:** rose from 32% to 57%, powered by AI/data domains.
-  **TypeScript:** moved from 12% to 37%, becoming a core web-stack skill.
-  **PHP:** fell from 30% to 17%, signaling shrinking demand.

Python Is Now the #1 Primary Language at 35%

Primary language leadership is clearer than adoption intent: developers rarely plan to switch their core stack.

Primary language share



Java at 30%

Comfort-zone effect

Adoption intent remains modest across all languages.

Python **6%**

TypeScript **5%**

Java **4%**

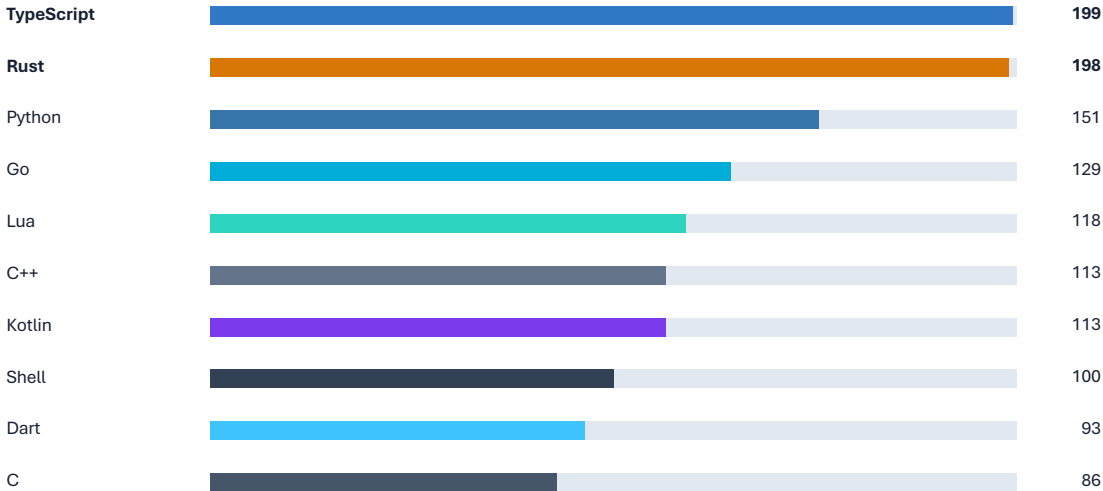
SQL **4%**

Implication: hiring plans should augment existing ecosystems, not assume rapid language switching.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

Developer enthusiasm clusters around type safety, modern tooling, and expressive systems for large codebases.

Satisfaction score, top 10 languages



Signal

199

TypeScript

198

Rust

Nearly tied at the top; both are associated with safety and modern developer experience.

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Language choice is domain-driven: teams select ecosystems that already dominate the work to be done.



What This Means for Your Technology Strategy in 2025

Translate language signals into hiring, platform investment, and retention priorities.

01

Invest in Python for AI/ML teams

It is the clear leader in AI/ML, data science, and data processing.

02

Treat TypeScript as a strategic frontend/full-stack skill

It leads UI at 62% and tops satisfaction at 199.

03

Use Rust and Go as infrastructure growth plays

They map to performance, concurrency, cloud-native tooling, and systems work.

04

Plan carefully around PHP and Ruby hiring

PHP fell from 30% to 17%; Ruby fell from 10% to 4%.

05

Optimize retention through modern DX

Satisfaction leaders point to type safety, tooling, and expressive systems.